

PashuRakshak (पशुरक्षक) - Animal Health & Outbreak Surveillance Platform

An End-to-End Clinical Screening, Geospatial Outbreak Sensing & Coordinated Epidemic Defense System

Problem Statement ID PS 26128 (Govt of Maharashtra & MSIS)	Platform Core Go 1.22+ / Triple SQLite WAL	Clinical Decision Engine Qwen2.5-VL & viggoVet-Reasoning-20B	Document Edition v4.1.0 Production Whitepaper
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1. Problem Statement Alignment (PS ID 26128)

Official Problem Statement 26128 • Government of Maharashtra & MSIS

Title: Efficient systems for early detection, prevention, and management of livestock diseases and animal health issues

Description: Livestock owners, field veterinarians, para-veterinary workers and government departments often lack a unified, realtime mechanism to identify emerging animal-health risks at the village, block and district levels. Disease symptoms may be reported late, diagnostic facilities may be distant, vaccination and treatment histories may be incomplete, and information from farms, veterinary dispensaries, laboratories, vaccination drives and surveillance programmes may remain fragmented. These gaps can delay containment, increase livestock mortality and productivity loss, raise the risk of zoonotic transmission, and affect farmers' incomes. The challenge is to create a practical system that enables early warning, rapid reporting, risk assessment, preventive action, referral and coordinated response.

The **PashuRakshak (पशुरक्षक)** platform is engineered as a direct solution to this mandate, uniting dairy farmers, pastoralists, licensed veterinary doctors, dispensaries, NGOs, and state animal husbandry authorities into an interconnected, real-time epidemiological grid.

2. Multi-Role Ecosystem & Geospatial Entity Mapping

To address fragmentation between field workers and clinical facilities, the platform partitions functionality across four primary user classes:

 Livestock Farmers & Dairy Owners (farmer) Registers farm name, village (गावाचे नाव), taluka, district, livestock types (Cattle, Buffalo, Goat, Sheep, Poultry, Swine), and herd size. Features livestock video/photo screening, real-time outbreak SOS radar, recovery reporting (POST /api/outbreaks/report-recovery), and nearby clinic directory.	 Companion Pet Parents (pet_owner) Registers address, city, district, and pincode. Features instant lesion & motion gait screening, broad-spectrum contagion alerts (e.g. Canine Parvovirus, Feline Panleukopenia), and one-tap doctor second-opinion requests.
 Licensed Veterinarians & Clinics (vet) Dedicated Doctor Decision Hub: Case Review Queue to inspect doubtful screenings, verify diagnoses, issue R prescriptions, manage live clinic operating hours, broadcast current field visiting locations, and post leave notices.	 NGOs & Government Dispensaries (ngo / gov) Real-time emergency inventory management for critical vaccines (FMD, Lumpy Skin, PPR), broad-spectrum antibiotics, antiseptics, and PPE kits with automated low-stock warnings and official circular broadcasting.

3. Dynamic Outbreak Sensing, Vet Consensus & Resolution Engine

 Rule 1: Veterinarian Consensus Trigger When 3 or more (≥ 3) licensed veterinarians confirm the same infectious disease (e.g. FMD, Lumpy Skin, PPR, Parvo, Anthrax) within a district, the system automatically flags a Verified Outbreak (status = 'active', severity = 'CRITICAL') and dispatches district-wide SOS alerts.	 Rule 2: Farmer Clustering Trigger When 3 or more (≥ 3) farmers in the same locality or taluka report matching clinical symptoms via multimodal scans or consult requests, an active outbreak cluster is automatically formed with actionable biosecurity protocols.
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Dynamic Outbreak Resolution Lifecycle

Recovery Reporting: Farmers or paravets submit recovery counts (POST /api/outbreaks/report-recovery). Active affected counts decrement dynamically.

Containment Clearance: When affected counts reach 0 or when veterinarians issue an official resolution certificate (POST /api/outbreaks/resolve), the outbreak transitions to status = 'resolved', severity = 'RESOLVED', the active red SOS flag is cleared, and an all-clear resolution notice is broadcast across the district.

4. End-to-End Architecture & Data Flow

 PASHURAKSHAK FRONTEND CLIENT (React 18 / Tailwind CSS) [Farmers: Herd Screening & Outbreak Radar] • [Vets: Decision Hub & Pashu Aadhaar Search] • [Dispensaries: Vaccine Stock Grid]		
REST API (JSON • Multipart/form-data • X-Auth-Token Header)		
 GO HIGH-PERFORMANCE BACKEND SERVICE (Go 1.22+ • CGO • net/http) Bcrypt Authentication • Outbreak Cluster Consensus Engine • FFmpeg 4-Frame Temporal Keyframe Extractor		
 Media Engine Content-Addressable Storage FFmpeg 4-Frame Video Ingestion Temporal Gait & Lesion Slices	 Clinical Decision Engine Visual Analysis (Qwen2.5-VL) Differential Reasoning (viggoVet) 4-Level Emergency Triage	 Surveillance & Consensus Vet Consensus ≥ 3 Trigger Farmer Cluster ≥ 3 Trigger Dynamic Recovery Clearance
Zero Lock-Contention WAL Persistence		
 user.db (v4) Users, Roles, Bcrypt Auth, 30-Day Sessions, Clinic Hours & Visiting Locations	 animal_health.db (v5) Pashu Aadhaar 12-Digit Tags, Health Passports, Doctor Lab Results, Screenings	 surveillance.db (v2) Outbreak Clusters, SOS Alerts, Doctor Review Queue, Vaccine Supplies, Circulars

5. Triple SQLite Schema Architecture & Data Flow Isolation

To ensure fault isolation and prevent surveillance queries from competing with clinic authentication, storage is split into three independent SQLite databases running in Write-Ahead Logging (WAL) mode:

Database & Engine	Table Name	Schema Structure & Key Indexes
user.db Schema v4 (WAL)	users	id (PK), name, email (UNIQUE), password_hash, role, district, clinic_name, is_available, visiting_location, leave_notes, created_at
	sessions	token (PK), user_id (INDEX), role, expires_at, created_at
	user_profiles	user_id (PK), farm_name, herd_size, village, taluka, district, clinic_hours, updated_at
	audit_logs	id (PK), user_id, email, event, ip_address, details, created_at
animal_health.db Schema v5 (WAL)	animals	id (PK), user_id, name, species, breed, age, sex, tag_number (12-digit Pashu Aadhaar), herd_size, village, taluka, district, photo_url, notes, weight, created_at
	health_screenings	id (PK), animal_id, user_id, media_url, symptoms_json, visual_analysis_json, assessment_json, urgency, district, taluka, farm_name, vet_review_requested, created_at
	clinic_test_results	id (PK), animal_id, user_id (owner), vet_id, vet_name, clinic_name, test_type, sample_date, test_parameters_json, interpretation, status ('Normal' 'Abnormal' 'Critical'), recommendation, created_at
	reminders	id (PK), user_id, animal_id, type, title, description, due_at, completed, created_at
surveillance.db Schema v2 (WAL)	outbreaks	id (PK), disease_name, species, district, taluka, village, farm_name, affected_count, severity, prevention_guide, status ('active' 'contained' 'resolved'), verified_by_vets, reported_by_farmers, resolution_notes, resolved_at, created_at
	notifications	id (PK), user_id, role_target ('all' 'farmer' 'vet' 'ngo'), district, title, message, outbreak_id, severity, is_sos, read, action_url, created_at
	vet_consultations	id (PK), screening_id, user_id, user_name, user_role, user_phone, location, vet_id, vet_name, animal_name, species, breed, media_url, symptoms_json, owner_notes, ai_diagnosis, ai_urgency, vet_diagnosis, vet_suggestion, vet_prescriptions, status ('pending' 'reviewed'), created_at
	inventories	id (PK), user_id, owner_name, org_type, district, item_name, category, quantity, unit, min_threshold, expiry_date, status, updated_at
	gov_advisories	id (PK), issuer, title, content, district, category, urgency, date_issued, created_at

6. Complete RESTful API Specifications

Method & Path	Headers & Params	Description & Payload Architecture
POST /api/auth/register	Content-Type: json	Registers account with role (farmer, pet_owner, vet, ngo) and geo fields.
POST /api/auth/login	Content-Type: json	Authenticates credentials with Bcrypt verification; issues 30-day session token.
POST /api/auth/profile	X-Auth-Token	Updates profile, farm parameters, or veterinarian duty status & leave notices.
GET /api/vets	?district=Pune	Returns directory of licensed veterinarians with real-time duty status and clinic hours.
POST /api/health-check/analyze	X-Auth-Token	Executes 4-frame temporal extraction, visual inspection, and differential assessment.
GET /api/outbreaks	?district=...	Lists active and resolved disease clusters across Maharashtra districts.
POST /api/outbreaks/report-recovery	Content-Type: json	Reports herd recovery. Automatically decrements affected counts and clears SOS.
POST /api/outbreaks/resolve	X-Auth-Token (Vet)	Doctor issues official clearance certificate to resolve outbreak.
GET /api/animals	?q=...&tag_number=...	Fetches user's animals or full regional directory with Pashu Aadhaar search (veterinarians).
GET /api/animals/tag/:tag	Public / Auth	Instant lookup of any livestock or companion animal by 12-digit Pashu Aadhaar ear tag.
GET /api/clinic-test-results	?animal_id=...	Confidential retrieval of doctor-issued lab test reports (owner & attending vets only).
POST /api/clinic-test-results	X-Auth-Token (Vet)	Veterinarian publishes official lab test report (CBC, Mastitis CMT, PCR).
GET /api/inventory	?district=...	Returns dynamic stock of vaccines, antibiotics, antiseptics, and PPE supplies.
PUT / DELETE /api/inventory	X-Auth-Token	Updates or deletes inventory stock (strictly restricted to the record creator).
GET /api/gov-advisories	?district=...	Returns official Maharashtra animal husbandry circulars and ring vaccination directives.

7. Clinical Verification Benchmarks across Livestock & Companion Animals

Clinical Scenario	Visual Analysis	Veterinary Differentials	Triage & Outbreak
Bovine Farm Cluster Baramati Dairy Farm, Cattle	Excessive ropy salivation, interdigital foot vesicles, oral erosions on tongue.	1. [HIGH] Foot-and-Mouth Disease (FMD / लाळ्या खुरकूत) 2. [MODERATE] Vesicular Stomatitis	CRITICAL SOS Verified Outbreak Active
Caprine Epidemic Sangamner Goat Herd	Mucopurulent ocular-nasal discharge, necrotic stomatitis, dyspnea.	1. [HIGH] Peste des Petits Ruminants (PPR) 2. [MODERATE] Contagious Caprine Pleuropneumonia	URGENT Ring Vaccine Deployed
Canine Motion Clip 10s video of puppy twitching	Rhythmic involuntary myoclonus (chewing-gum seizures), nasal hyperkeratosis.	1. [HIGH] Canine Distemper Encephalomyelitis 2. [MODERATE] Idiopathic Epilepsy	EMERGENCY Doctor Consult Sent
Healed Dairy Herd Malegaon Budruk, Day 14 Follow-up	Complete re-epithelialization of oral lesions, normal gait, feeding resumed.	1. [RESOLVED] Post-FMD Clinical Recovery 2. Complete tissue healing confirmed.	RESOLVED SOS Cleared • Open